

Classroom Live: Live Bilingual Classroom Translation

ADI205/501 - AI Augmented Professional Practice \ **Status:** Development report based on the current source code, team observations, and recorded demonstrations. The final live classroom demonstration is pending.

1. Objective and team

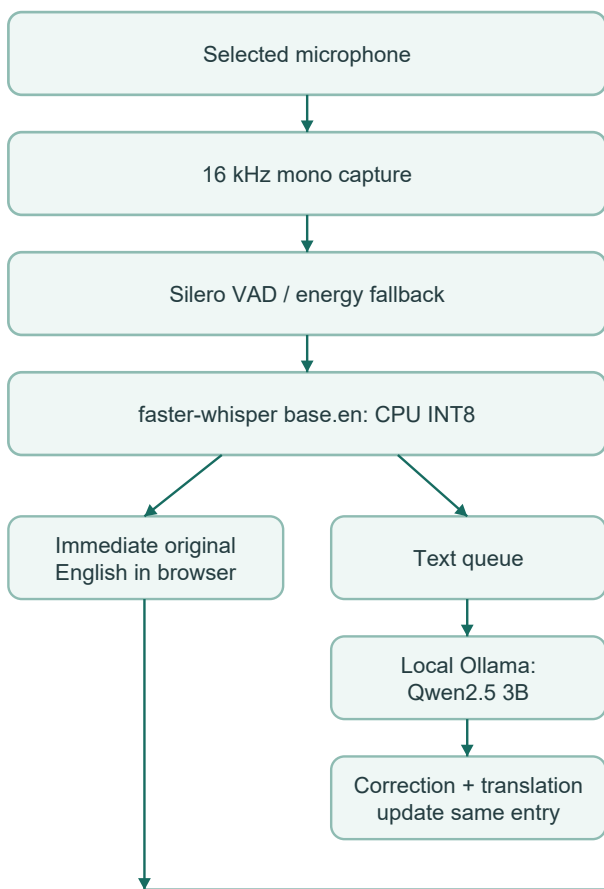
Classroom Live helps students follow English lectures by displaying original English transcription, corrected English, and a Chinese or Vietnamese translation. It also processes recorded audio/video and exports text for later review. Privacy and affordability guided the design: audio and text inference run locally rather than being sent to an external API. Initial software and model downloads require internet access; local processing avoids per-use API fees but still uses the computer's resources.

Member	Contribution
Jerry Chan	Implemented core functionality: transcription, the basic webpage, and Ollama correction and translation.
Owen Liu	Compiled the technical decisions and their justification.
Yiyang Ge	Updated the UI, extended recorded transcription, and changed live output to display English before correction and translation.
Quang	Contribution to be confirmed.

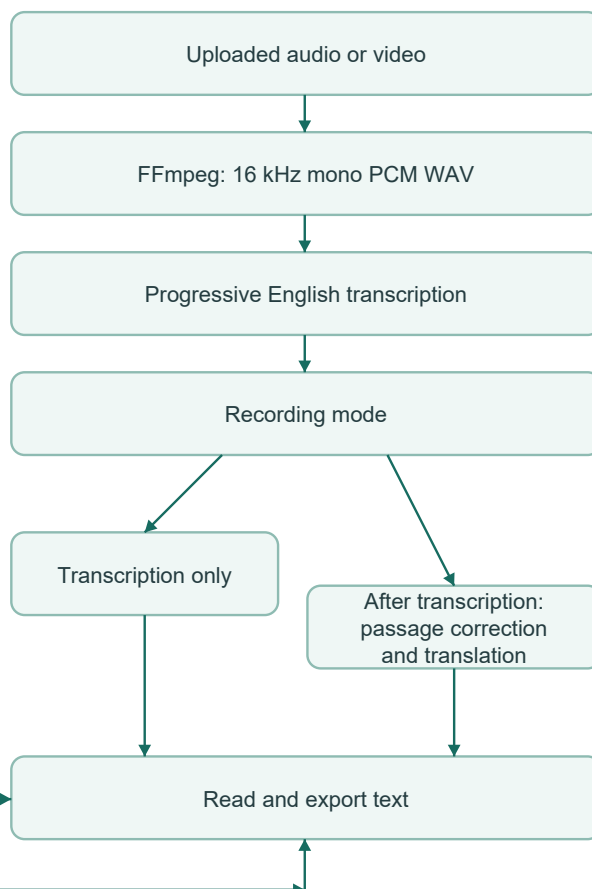
The assignment requires live microphone input, incremental bilingual output, correction, engineering comparisons, testing evidence, and approximately three seconds or less of end-to-end delay. The prototype provides the main workflows, but the full corrected translation does not meet that timing target in the reported observations.

2. Architecture and operation

LIVE MICROPHONE



RECORDED MEDIA



FastAPI serves the browser interface and coordinates the pipeline. Live speech recognition and correction/translation run in separate workers. Recognized English is published through WebSockets immediately after recognition; the slower model updates the corresponding transcript entry later. Original English remains available, and corrected English is shown when it differs. The live English panel joins chunks into growing sentences, using approximate punctuation boundaries.

Recorded media has its own workspace, settings, results, and exports. FFmpeg extracts and normalizes audio before recognition. English appears progressively through streamed HTTP responses. In translation mode, the application completes recognition before translating grouped passages with neighboring context. Users can select approximate passage lengths, stop processing, and retain partial results. Transcription-only mode skips correction and translation. Stop is cooperative: an operation already running may finish before cancellation is acknowledged.

3. Engineering decisions

The group used Perplexity-assisted research to investigate alternatives. The supplied research export discusses Whisper implementations and cloud transcription services. These were researched options, not a controlled benchmark of every alternative.

Stage	Selected implementation and rationale	Alternatives and tradeoffs
Audio capture	sounddevice with user selection from detected input devices. The built-in microphone avoids extra equipment and suits a portable prototype.	USB, wireless/lapel, and virtual inputs can be used if detected. External microphones may improve distant speech capture but add cost and setup; they were not evaluated here.
Noise handling	No dedicated noise-removal filter. Silero VAD identifies speech, with an RMS-energy fallback if unavailable. This keeps the initial pipeline simple.	Noise gates, RNNoise, and device/OS suppression were considered. They require tuning or additional integration and may affect quiet speech. VAD is speech detection, not noise removal; noisy-classroom testing remains pending.
Transcription	Local faster-whisper, using English base.en with CPU INT8 and six configured CPU threads. It integrates directly into Python and limits model size and computing demand.	Original Whisper, whisper.cpp, larger Whisper models, cloud APIs, and OS/meeting transcription were considered. The group favored Python integration and local control; larger or cloud models would need separate accuracy, latency, and resource evaluation.
Correction	Qwen2.5 3B through local Ollama corrects English in the same request that translates it. Python compares original and corrected words for display.	Rule-based correction is simpler but less contextual. A cloud language model introduces external text processing and possible charges. The local model can still miss errors or change meaning.
Translation	Reuse Qwen2.5 3B for Chinese or Vietnamese, reducing the number of models and integration steps.	A separate local translation model or cloud service could be evaluated for quality and speed. The current joint correction/translation call is the main observed bottleneck.
UI	HTML, CSS, and JavaScript with FastAPI/Uvicorn; WebSockets deliver live updates. A browser interface fits the existing Python backend and supports flexible layouts.	A Python desktop GUI adds a different UI framework; native macOS development would not suit the Windows demo device. Browser rendering does not eliminate inference delay.
Media formats	Direct FFmpeg conversion to 16 kHz mono, 16-bit PCM WAV, then NumPy float32 samples for recognition. This gives explicit control over normalization.	Pydub adds a wrapper; direct faster-whisper decoding could simplify the pipeline; WAV-only input burdens users; cloud conversion sends recordings outside the computer.

The faster-whisper documentation publishes implementation benchmarks and supports CPU INT8, but its results are not measurements of this project or Surface Pro 9. [Faster-whisper documentation](#).

Groq Whisper Large v3 was a researched cloud transcription alternative. Its documentation lists usage pricing and upload limits, illustrating the constraints the group wanted to avoid. It is not the model used for Chinese/Vietnamese correction and translation. [Groq model documentation](#).

Tools and versions

The repository specifies the following **minimum package versions**, not a verified record of the demo environment:

Packages	Declared minimum versions
FastAPI / Uvicorn / websockets	0.110.0 / 0.28.0 / 12.0
sounddevice / NumPy	0.4.6 / 1.26.0
faster-whisper / ONNX Runtime / silero-vad	1.0.0 / 1.17.0 / 5.1
requests / python-multipart	2.31.0 / 0.0.9

Configured model identifiers are `base.en` and `qwen2.5:3b`. Exact installed Python, package, FFmpeg, and Ollama versions were not recorded in the supplied demo evidence and should be captured before submission.

4. Development and UI improvements

Speed was the largest development problem. The team observed that Ollama correction and translation took substantially longer than English recognition. Publishing English first gave users useful feedback sooner, while preserving later correction and translation. This improved perceived responsiveness without establishing a reduction in final translation latency.

The interface also evolved from a shared live/recorded display to separate workspaces. Earlier recordings could fill the live subtitle area with a long block of text. Separate recording results, paragraph grouping, timestamps, explicit controls, and progress made this workflow easier to follow.

The following screenshots document development stages, not a complete view of the latest live layout. Their displayed processing counters are not validated end-to-end timing measurements.

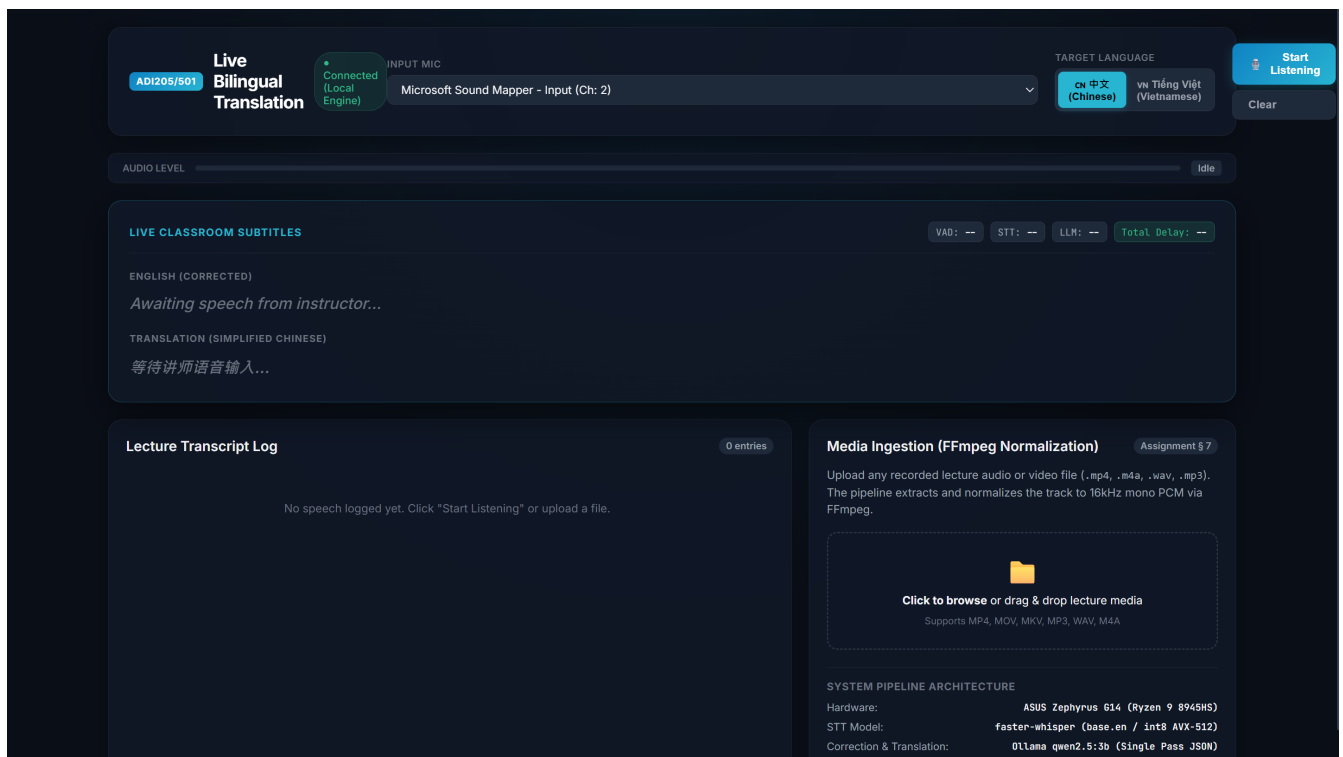


Figure 1. Original dark interface before the UI redesign. The hardware label belongs to that earlier interface; the reported demo device was a Surface Pro 9.

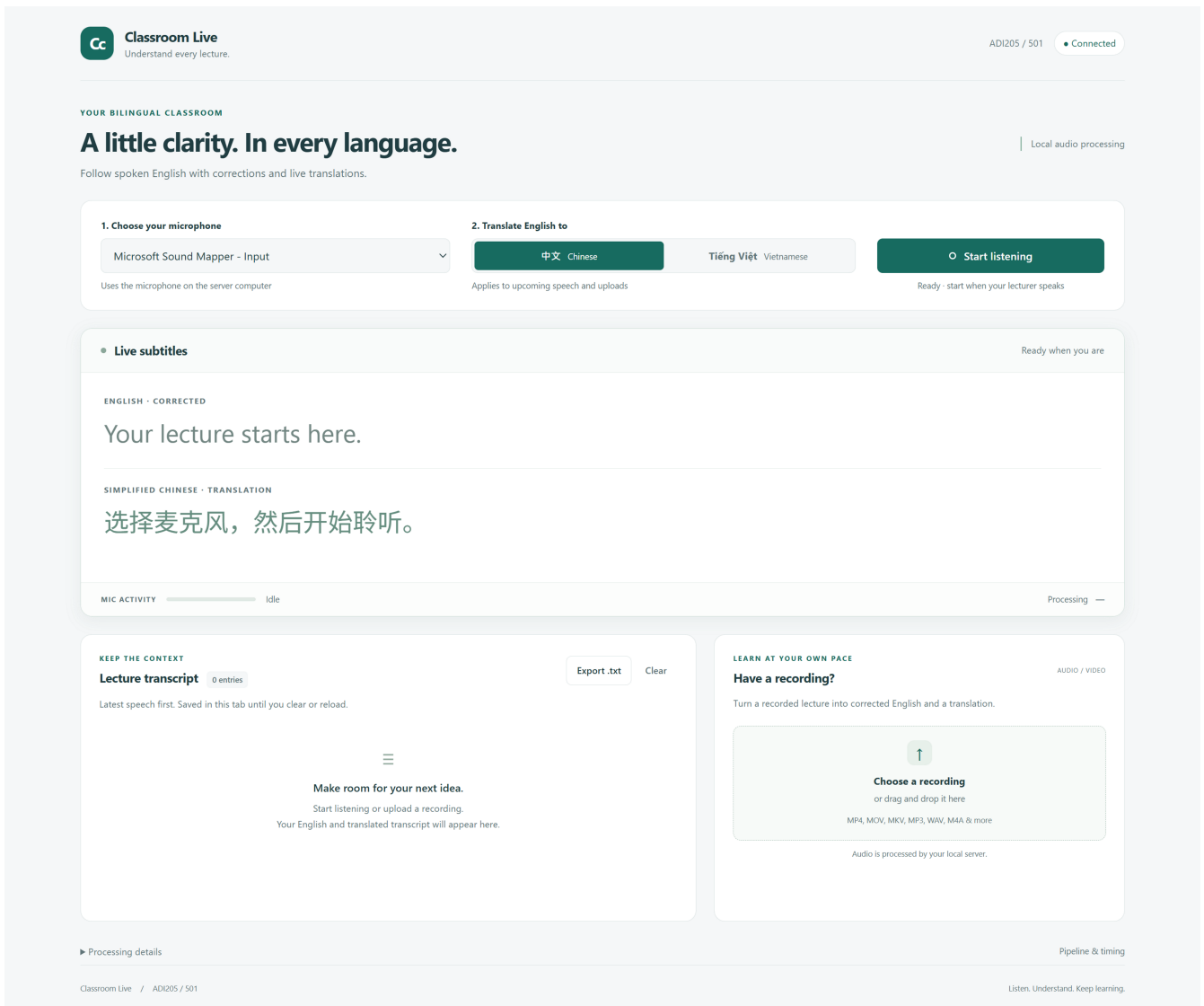


Figure 2. Initial redesign with clearer microphone/language controls, larger text, and transcript export.

Classroom Live

Understand every lecture.

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Connected

YOUR BILINGUAL CLASSROOM

A little clarity. In every language.

Local audio processing

Follow spoken English with corrections and live translations.

1. Choose your microphone

Microsoft Sound Mapper - Input

Uses the microphone on the server computer

2. Translate English to

中文 Chinese

Tiếng Việt Vietnamese

Start listening

Ready - start when your lecturer speaks

Live subtitles

Ready when you are

ENGLISH - CORRECTED

Welcome to the Assemblymen with the New York State Legislature. We've been in the Assembly for a while over ten years at this point. He's now in the, he's in the minority, the Republican aspect of the legislature, which as I've talked about in class, and this is for our American government and civic responsibility class. As you know, it's a, you know, two-house legislature and the governor of Democrat, and the assembly is primarily Democrat, and the Senate is primarily Democrat. So Carl has always been cheery and willing to accept whatever comes his way. One of the great things about him, but he has his assistant assembly minority leader, which he's in the number three position in the assembly minority, which is pretty good. He also, though, has, prior to becoming an assemblyman, Carl was on the town board for the town of Deer Park, and we talked about town government, and then he was the supervisor for the town of Deer Park, an institute at certain cost savings mechanisms and things of that nature helped the town run well. And since then, he's been unopposed in a number of elections. Are you unopposed this year? This year, yeah. Again, it's all about a little, probably a few days ago. Yeah, all the auditions had to be in, so unopposed again, no one's afraid to run against him. And on the bile sec here. It did work good. Well, the bile sec was our former state senator and assemblyman who actually has a lawyer at some point. So anyway, Assemblyman Brabnack, we're here to talk about the New York State Legislature and also what we're covering this week in our course is lobbying, persuasion, and political influence. And I'm sure that as a member of the assembly, Assemblyman Brabnack, I'm sure has dealt with lobbyists, and interest groups and things of that nature. So he has free reign to talk about whatever he wants. I hope you all have great questions.

SIMPLIFIED CHINESE - TRANSLATION

Offline (Simplified Chinese)

Original English

Welcome to the Assemblymen with the New York State Legislature. We've been in the Assembly for a while over ten years at this point. He's now in the, he's in the minority, the Republican aspect of the legislature, which as I've talked about in class, and this is for our American government and civic responsibility class. As you know, it's a, you know, two-house legislature and the governor of Democrat, and the assembly is primarily Democrat, and the Senate is primarily Democrat. So Carl has always been cheery and willing to accept whatever comes his way. One of the great things about him, but he has his assistant assembly minority leader, which he's in the number three position in the assembly minority, which is pretty good. He also, though, has, prior to becoming an assemblyman, Carl was on the town board for the town of Deer Park, and we talked about town government, and then he was the supervisor for the town of Deer Park, an institute at certain cost savings mechanisms and things of that nature helped the town run well. And since then, he's been unopposed in a number of elections. Are you unopposed this year? This year, yeah. Again, it's all about a little, probably a few days ago. Yeah, all the auditions had to be in, so unopposed again, no one's afraid to run against him. And on the bile sec here. It did work good. Well, the bile sec was our former state senator and assemblyman who actually has a lawyer at some point. So anyway, Assemblyman Brabnack, we're here to talk about the New York State Legislature and also what we're covering this week in our course is lobbying, persuasion, and political influence. And I'm sure that as a member of the assembly, Assemblyman Brabnack, I'm sure has dealt with lobbyists, and interest groups and things of that nature. So he has free reign to talk about whatever he wants. I hope you all have great questions.

No corrections

MIC ACTIVITY

Idle

Processing 37.43 s

KEEP THE CONTEXT

Lecture transcript

5 entries

Export .txt

Clear

Latest speech first. Saved in this tab until you clear or reload.

1:00:25 PM - Chinese - 37.43 s processing

Welcome to the Assemblymen with the New York State Legislature. We've been in the Assembly for a while over ten years at this point. He's now in the, he's in the minority, the Republican aspect of the legislature, which as I've talked about in class, and this is for our American government and civic responsibility class. As you know, it's a, you know, two-house legislature and the governor of Democrat, and the assembly is primarily Democrat, and the Senate is primarily Democrat. So Carl has always been cheery and willing to accept whatever comes his way. One of the great things about him, but he has his assistant assembly minority leader, which he's in the number three position in the assembly minority, which is pretty good. He also, though, has, prior to becoming an assemblyman, Carl was on the town board for the town of Deer Park, and we talked about town government, and then he was the supervisor for the town of Deer Park, an institute at certain cost savings mechanisms and things of that nature helped the town run well. And since then, he's been unopposed in a number of elections. Are you unopposed this year? This year, yeah. Again.

LEARN AT YOUR OWN PACE

Have a recording?

Turn a recorded lecture into corrected English and a translation.

Choose a recording

or drag and drop it here

MP4, MOV, MKV, MP3, WAV, M4A & more

Speech assemblyman cut.m4a

Complete - added to your transcript

Audio is processed by your local server.

Processing details

Pipeline & timing

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Listen. Understand. Keep learning.

Figure 3. Earlier problem: recorded text occupied the live subtitle area without paragraph separation. Translation was unavailable in this historical example.

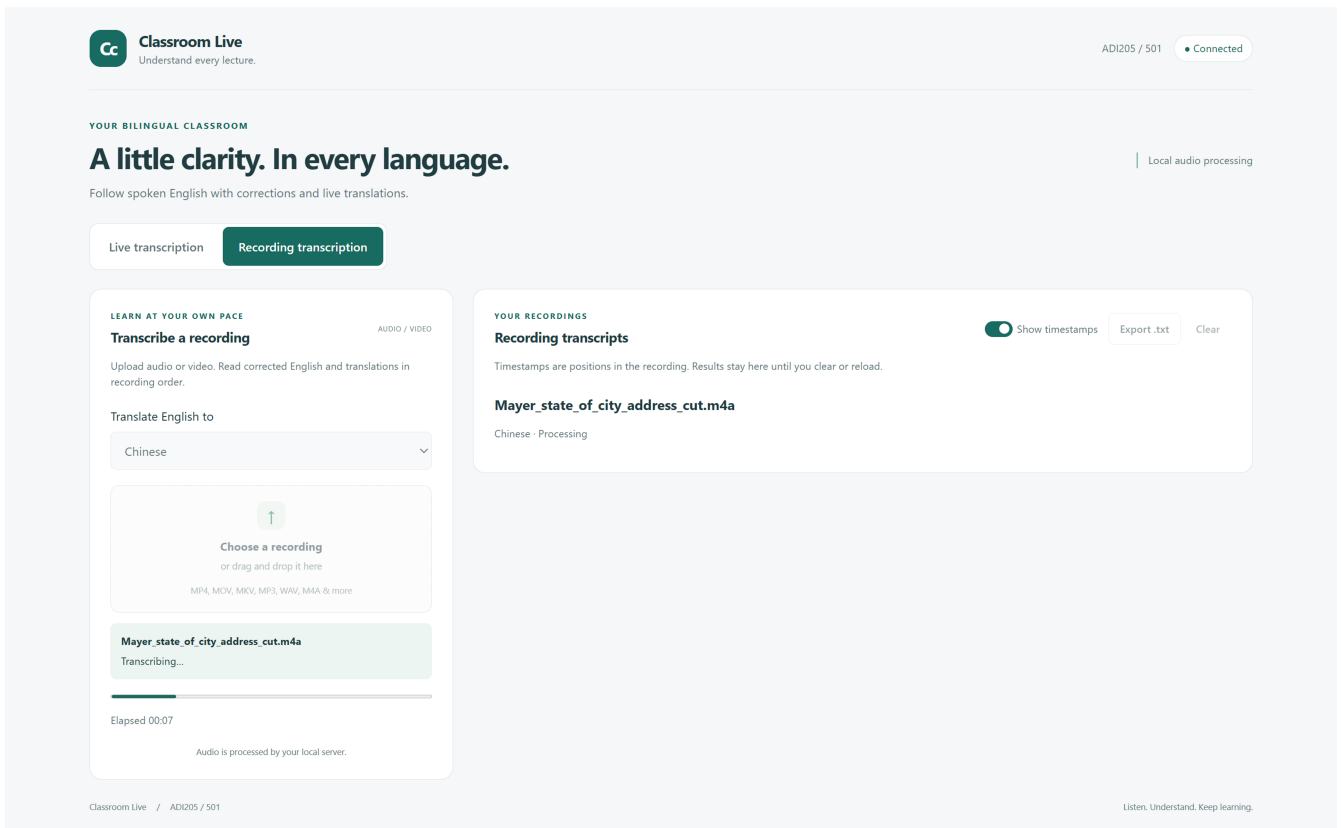


Figure 4. A dedicated recording workspace separates uploaded-media processing from live transcription.

Cc

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YOUR BILINGUAL CLASSROOM

A little clarity. In every language.

Local audio processing

Follow spoken English with corrections and live translations.

Live transcription

Recording transcription

LEARN AT YOUR OWN PACE

Transcribe a recording

AUDIO / VIDEO

Upload audio or video. Read draft English first, then corrections and translations as they finish.

Translate English to

Chinese

Translation segment length

Auto - about 60 seconds

Lengths are approximate. Passages end at sentence boundaries where possible and use neighboring text for context.

Choose a recording

or drag and drop it here

MP4, MOV, MKV, MP3, WAV, M4A & more

Speech_assblyman_cut.m4a

Start transcription

Stop processing

Speech_assblyman_cut.m4a

Transcript ready · Translating paragraph 1 of 2

Elapsed 00:37

Audio is processed by your local server.

YOUR RECORDINGS

Recording transcripts

Show timestamps

Export .txt

Clear

Timestamps are positions in the recording. Results stay here until you clear or reload.

Speech_assblyman_cut.m4a

Chinese · Transcript ready · Translating paragraph 1 of 2

00:00-01:08

ENGLISH · DRAFT (UNCORRECTED)

Welcome to the Assemblymen with the New York State Legislature. We've been in the Assembly for a while over ten years at this point. He's now in the, he's in the minority, the Republican aspect of the legislature, which as I've talked about in class, and this is for our American government and civic responsibility class. As you know, it's a, you know, two-house legislature and the governor of Democrat, and the assembly is primarily Democrat, and the Senate is primarily Democrat. So Carl has always been cheery and willing to accept whatever comes his way. One of the great things about him, but he has his assistant assembly minority leader, which he's in the number three position in the assembly minority, which is pretty good. He also, though, has, prior to becoming an assemblyman, Carl was on the town board for the town of Deer Park, and we talked about town government, and then he was the supervisor for the town of Deer Park, an institute at certain cost savings mechanisms and things of that nature helped the town run well.

SIMPLIFIED CHINESE · TRANSLATION

Translating...

01:08-01:59

ENGLISH · DRAFT (UNCORRECTED)

And since then, he's been unopposed in a number of elections. Are you unopposed this year? This year, yeah. Again, it's all about a little, probably a few days ago. Yeah, all the auditions had to be in, so unopposed again, no one's afraid to run against him. And on the bile sec here. It did work good. Well, the bile sec was our former state senator and assemblyman who actually has a lawyer at some point. So anyway, Assemblyman Brabnack, we're here to talk about the New York State Legislature and also what we're covering this week in our course is lobbying, persuasion, and political influence. And I'm sure that as a member of the assembly, Assemblyman Brabnack, I'm sure has dealt with lobbyists, and interest groups and things of that nature. So he has free reign to talk about whatever he wants. I hope you all have great questions.

SIMPLIFIED CHINESE · TRANSLATION

Translation pending...

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Listen. Understand. Keep learning.

Figure 5. Later recording workflow adds passage-length selection, Start/Stop controls, timestamps, and English output while translation remains pending. This is the sixth supplied screenshot; the fifth supplied screenshot is intentionally excluded.

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5. Testing and results

Environment and procedure

The recorded demonstration ran on a **Surface Pro 9, Windows 11, Intel Core i7-1255U (2.60 GHz), 16 GB RAM, and Intel Iris Xe integrated graphics**. It used the built-in microphone in a quiet environment. The team observed live English and translation output, reviewed exported text, and tested recorded-media formats. The final live demonstration and realistic classroom-noise testing have not yet been completed.

Check	Reported result	Evidence boundary
Speech to original English	Approximately 2-3 seconds	Informal estimate from speaking, not instrumented statistics.
Speech to corrected translation	Approximately 7-8 seconds total	Informal estimate from speaking, not an additional 7-8 seconds after English. Exceeds the assignment target.
Recorded formats	MKV, MOV, MP4, FLAC, WAV, MP3, M4A	No failures reported in tested transcription, translation, and export; does not establish compatibility with every codec/file.
Target languages	Mostly Chinese; Vietnamese tried once	Vietnamese quality has not been thoroughly evaluated.
Recognition and translation quality	Clearer spoken English generally produced better results	Qualitative observation; no reference transcript or numerical accuracy score.
Classroom noise	Not yet evaluated. Can be avoided with wireless microphone.	Quiet-environment results cannot establish classroom performance.

The code's displayed latency adds recognition time, model time, and a fixed 10 ms. It omits audio accumulation and queue/display delays, so it cannot verify the full speech-to-display target. Current live VAD permits chunks up to 4.5 seconds, further limiting any universal sub-three-second claim.

Accuracy example

The updated live demonstration is accompanied by this exported example:

Session timestamp	Original English	Corrected English	Chinese translation
00:12	transform your life.	transform your life.	改变你的生活。
00:16	Success is something you attract by the person you become.	Success is something you attract by the person you become.	成功是你成为的人吸引来的。
00:18	Jim Rohn.	Jim Rohn.	Jim Rohn.

This sample provides a clearer example of useful output: the first translation is natural, the second conveys the main idea but is somewhat literal, and the proper name is preserved in English. These are qualitative observations rather than a numerical accuracy assessment. The timestamps indicate session positions, not processing delays.

All three excerpts have identical original and corrected English, with no corrections listed.

Regression checks and evidence

The repository contains Python tests for streaming, cancellation, error handling, and live publication, plus browser checks for workspace isolation, progress, exports, layout, and safe text rendering. Development history reports passing checks, but these use simulated inference and are not proof of real-model accuracy or microphone performance. They were not rerun while writing this report.

Evidence included in the project:

- Updated live demonstration video (demo/Demo_live_trans_new.mp4)
- Recorded-media demonstration video (demo/Recorded_transcription_demo.mp4)
- Updated exported live transcript (demo/live-transcript-2026-09-13.txt)
- Regression-test instructions and scope (tests/README.md)
- Development screenshots in Section 4.

6. Running the application

These instructions use a standard Python virtual environment on Windows; Conda is not required. Install a compatible 64-bit Python distribution, [Ollama](#), and [FFmpeg](#). Make `python`, `ollama`, and `ffmpeg` available in the terminal. FFmpeg is required by this application's recording conversion step.

From the project root in PowerShell:

```
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
ffmpeg -version
ollama pull qwen2.5:3b
```

Ensure Ollama is running at `http://localhost:11434`. If its desktop application is not serving it, run `ollama serve` in a separate terminal. Then launch:

```
.\.venv\Scripts\python.exe main.py
```

Open **`http://localhost:8000`** after initialization finishes. First startup may download the Whisper model. No external API key is required. In Windows microphone privacy settings, allow microphone access for desktop applications. The Python server captures the microphone on its own computer.

For live use, select the microphone and target language, then start listening. Read English as it appears and the correction/translation when available; stop listening and export the transcript as needed. For recordings, choose the recording tab, select a mode and file, adjust translation settings if applicable, and press Start. Stop preserves partial results; export before clearing or reloading the page.

If there is no input, check microphone selection and permissions. If translation fails, check Ollama and confirm `qwen2.5:3b` appears in `ollama list`. If recording conversion fails, check FFmpeg and whether the file contains a decodable audio track. Restart the server and refresh the browser after code changes.

To run backend checks, install the API test dependency and use:

```
.\.venv\Scripts\python.exe -m pip install httpx
.\.venv\Scripts\python.exe -m unittest discover -s tests -v
```

Browser checks additionally require Node.js, Playwright, and Microsoft Edge; see the linked testing instructions. Capture the actual environment for reproducibility with `python --version`, the environment's `python -m pip freeze`, `ffmpeg -version`, and `ollama --version`.

7. Limitations and next steps

The principal limitation is the 7-8-second observed corrected-translation delay. English-first output helps users follow speech, but does not satisfy the full bilingual timing requirement. Secondly, sentence splitting can remove context.

Next steps are to develop a mechanism that well handles context preservation during live transcription instead of only in recorded transcription. An optional paid cloud correction/translation service could be compared with local inference for speed and quality. Any benefit must be weighed against charges, internet dependence, and sending text outside the computer.

The prototype establishes a local live and recorded transcription workflow. Its main engineering improvement is separating immediate English feedback from slower correction and translation. Current evidence supports functional progress and identifies concrete accuracy and latency limitations that largely depends on the hardware given it is all local based and tested demo is not on a strong computer.

References

1. Zheng Qu, *ADI205/501 Group Project: Live Bilingual Classroom Translation Application*, August 27, 2026. Assignment PDF supplied with the repository.
2. Team technical-decision notes and interview responses supplied by Yiyang Ge.
3. Perplexity research export, *any good and free model for transcription*, supplied by the team. Used as evidence of alternatives researched, not a project benchmark.
4. SYSTRAN: faster-whisper.
5. Ollama: Qwen2.5 3B.
6. Silero VAD documentation.
7. FFmpeg documentation.
8. Groq: Whisper Large v3.

Official web references checked during report preparation in September 2026. Vendor claims are not substituted for project measurements.